

12(a)	quantum of energy	M1
	of electromagnetic radiation	A1
12(b)(i)	energy = hc / λ or energy = hf and $f = c / \lambda$	C1
	$0.57 \times 10^6 \times 1.60 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 2.2 \times 10^{-12} \text{ m}$	A1

Question	Answer	Marks
12(b)(ii)	$p = h / \lambda$	C1
	$= (6.63 \times 10^{-34}) / (2.2 \times 10^{-12})$ $= 3.0 \times 10^{-22} \text{ N s}$	A1
	or	
	$p = E / c$	(C1)
	$= (0.57 \times 10^6 \times 1.60 \times 10^{-19}) / (3.00 \times 10^8)$ $= 3.0 \times 10^{-22} \text{ N s}$	(A1)
12(c)(i)	mass (of Sm-157 nucleus) = $157 \times 1.66 \times 10^{-27}$ or mass (of Sm-157 nucleus) = $0.157 / (6.02 \times 10^{23})$	C1
	recoil speed = $(3.00 \times 10^{-22}) / (157 \times 1.66 \times 10^{-27})$ $= 1.2 \times 10^3 \text{ m s}^{-1}$	A1
12(c)(ii)	$(1.2 \times) 10^3 \text{ m s}^{-1}$ is <u>much</u> less than $(3.0 \times) 10^8 \text{ m s}^{-1}$	B1